Urban Planning Companion

> Consolidated reference of all indicators, sample readings, 2030 targets, map markers, and calculation methodologies currently displayed in the demo. Data as of **Q1 2026** (simulated).

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1. Diagnostics and Forecasting

Page route: `/diagnostics_and_forecasting` **Full name:** Diagnostics and Forecasting

This page monitors Saudi Arabia's urban performance across traffic flow, housing demand, idle land, and asset competitiveness — all benchmarked against Vision 2030 targets.

1.1 Flow Agent

> Forecasts citizen movement patterns and changing demand for services.

1.1.1 24H Commute Efficiency Index (`flw_1`)

Description: AI detects non-recurrent congestion anomalies using real-time computer vision feeds. The index measures how close a city's traffic flow is to optimal (free-flow) conditions, factoring in accidents, weather, and special events via dynamic penalty modifiers.

Metric	Value
Current Index (Q1 2026)	**78** / 100
2030 Target	**92** / 100
Gap	+14 pts
Critical Alerts	12
Warnings	34
Active Cameras	142

Calculation formula:

```
INDEX = 100 × (V_ACTUAL / V_BASE) × C_MOD
```

- **V_ACTUAL** — Current segment average velocity via computer vision and IoT (real-time feeds).
- **V_BASE** — Expected non-peak steady-state velocity (13:00–16:00 historical average).
- **C_MOD** — Dynamic penalty factor (0.6–1.0) for accidents, weather, and non-cyclical events.

Interpretation matrix:

Index Range	Status	Action
90–100	Optimal	Maintain
70–89	Moderate	Monitor
50–69	Degraded	Intervene
0–49	Critical	Emergency response

Map markers (12 Riyadh locations):

#	Location	Severity	Extra Delay	Speed Drop
1	King Fahd Rd	CRITICAL	+45m	-81.5%
2	Olaya St	HIGH	+32m	-62.5%
3	Northern Ring	CRITICAL	+50m	-77.7%
4	King Abdullah Rd	HIGH	+28m	-64.0%
5	Khureis Rd	CRITICAL	+40m	-82.5%
6	Makkah Rd	HIGH	+25m	-63.3%
7	Takhassusi St	CRITICAL	+38m	-82.2%
8	Eastern Ring	HIGH	+20m	-61.1%
9	Abu Bakr Rd	CRITICAL	+42m	-85.7%
10	Prince Turki	HIGH	+22m	-64.0%
11	King Salman Rd	HIGH	+18m	-50.0%
12	Dirab Rd	CRITICAL	+35m	-76.9%

1.1.2 City Congestion Ranking (`flw_4`)

Description: Real-time national traffic congestion index by city. Critical threshold: 60+.

Metric	Value
Cities Tracked	8
Critical (≥60)	4
High	2

City rankings:

Rank	City	Region	Congestion Index	Avg Delay	Critical Roads	Affected Pop.	Status
1	Riyadh	Riyadh	78	+35 min	12	2.3M	CRITICAL
2	Jeddah	Makkah	72	+28 min	8	1.5M	CRITICAL
3	Makkah	Makkah	68	+32 min	6	1.2M	HIGH
4	Dammam	Eastern	62	+22 min	5	850K	HIGH
5	Madinah	Madinah	58	+18 min	4	620K	WARNING
6	Khobar	Eastern	55	+15 min	3	480K	WARNING
7	Tabuk	Tabuk	48	+12 min	2	320K	MODERATE
8	Buraidah	Qassim	45	+10 min	2	280K	MODERATE

1.2 Demand Forecaster Agent

> Forecasts future needs for housing, infrastructure (roads), and amenities across regions.

1.2.1 Housing Demand Forecast (`dmd_1`)

Description: Housing shortfall by 2030 across 6 regions. Uses NHC Sakani data + population growth models. 2030 Target: 70% homeownership (*real). Current: 63%. Deficit = units to close gap.

Metric	Value
By-2030 Deficit	**-153K** units
Current Homeownership (Q1 2026)	**63%**
2030 Target	**70%** (*real)

Map markers (6 regions):

Region	Deficit	Current Ownership	Target	Population
Riyadh	-35,000	55%	70%	8.6M
Makkah	-42,000	52%	70%	9.0M
Eastern Province	-28,000	65%	70%	5.1M
Madinah	-18,000	60%	70%	2.2M
Asir	-15,000	72%	70%	2.3M
Qassim	-15,000	71%	70%	1.5M

1.2.2 Road Network Expansion (`dmd_2`)

Description: Based on National Transport Strategy (2021–2030) and giga-project corridor studies. Current: 76K KM. Target: 100K KM by 2030.

Metric	Value
Current Network (Q1 2026)	**76K KM**
By-2030 Gap	**24K KM**
2030 Target	**100K KM**

Map markers (6 road projects):

Project	Length	Status	Target Completion
Riyadh–NEOM Corridor	4,200 KM	PLANNED	2029
Jeddah–KAEC Expressway	1,800 KM	IN PROGRESS	2028
Eastern Province Ring Road	3,500 KM	DESIGN	2030
Madinah–Yanbu Highway	2,100 KM	IN PROGRESS (45%)	2027
Abha–Soudah Tourism Rd	800 KM	DESIGN	2028
Qassim Agricultural Route	1,600 KM	PLANNED	2029

1.3 Idle Land Agent

> Identifies and maps underdeveloped or vacant parcels for redevelopment potential.

1.3.1 White Land Activation Rate (`idl_1`)

Description: AI scans MOMRAH White Land Registry + satellite imagery to pinpoint idle parcels. Explains why each plot is undeveloped and recommends use. Applies White Land Tax (2.5% annual levy) (*real — approved by Saudi Cabinet, applies to plots >5,000 sqm).

Metric	Value
Current Activation Rate (Q1 2026)	**38%**
2030 Target	**65%**
AI-Identified Spots	**12**

The Three Primary Criteria

For a piece of land to be classified as "White," it must check these three boxes simultaneously:

Vacancy: The land must be undeveloped or idle. There is no permanent structure on it that serves its intended purpose (simply putting a fence around it does not count as "development").

Urban Boundary: The land must fall within the Urban Scope (the geographical limits of a city as defined by official maps). Land outside these borders is not subject to the white land fee.

Service Readiness: This is the most critical technical detail. The land must have access to essential infrastructure, such as paved roads, electricity, water, and sewage. The logic is that if the government has already invested in the infrastructure around your land, you shouldn't be "hoarding" it while others need housing.

Map markers (12 spots across 6 regions):

#	Location	Area	Years Idle	Severity
1	Al Yasmin Block 4A, Riyadh	14,500 SQM	4 yrs	CRITICAL
2	Al Malqa Apex, Riyadh	32,000 SQM	6 yrs	CRITICAL
3	Khashm Al Aan South, Riyadh	22,000 SQM	3 yrs	HIGH
4	Al Shara'i Plateau, Makkah	48,000 SQM	5 yrs	CRITICAL
5	North Jeddah Corniche Strip	18,000 SQM	4 yrs	HIGH
6	Dhahran Valley West, Eastern	28,000 SQM	3 yrs	HIGH
7	Jubail 2 Southern Buffer, Eastern	35,000 SQM	5 yrs	HIGH
8	KEC Phase 2 North, Madinah	20,000 SQM	4 yrs	HIGH
9	Al Manar Commercial, Madinah	12,000 SQM	3 yrs	HIGH
10	Soudah Gateway Parcel, Asir	15,000 SQM	2 yrs	HIGH
11	Buraidah North University Belt, Qassim	25,000 SQM	4 yrs	HIGH
12	Al Rass Agri-Tech Hub, Qassim	18,000 SQM	3 yrs	HIGH

1.3.2 Urban Land Utilization Index (`idl_2`)

Description: AI analyzes satellite footprint, infrastructure density, and zoning to identify underutilized zones. Scores impact per zone and recommends development action.

Metric	Value
Current Index (Q1 2026)	**58** / 100
2030 Target	**80** / 100

Metric	Value
Zones Flagged	**13**

The Urban Land Utilization Index (ULUI) in the context of Saudi Vision 2030 (specifically within MOMRAH's tracking) is a key performance indicator that measures the efficiency of spatial development.

Based on current 2026 data and the specific numbers you cited (Current: 58, Target: 80), this index most likely refers to the Percentage of Urban Clusters Hosting Parks and Green Spaces within an 800-meter radius (the "10-minute walk" rule).

How is it Calculated?

The index is a spatial ratio that evaluates how much of the "built-up" urban area is effectively utilized for public benefit (like parks, services, or optimized housing) compared to the total planned urban land.

The Formula:

$$ ULUI = \left(\frac{\text{Area of Urban Clusters with Active Services/Parks}}{\text{Total Developed Urban Area}} \right) \times 100 $$

Numerator: Land that is not just "built" but "activated" (e.g., has a park, is mixed-use, or is fully integrated into the utility grid).

Denominator: The total footprint of land designated as "urban" within a city's boundaries.

What do the numbers mean?

Current (58): This means that as of late 2024/early 2025, approximately 58% of urban neighborhoods in Saudi Arabia have met the "humanized" city criteria (e.g., access to a park within 800 meters or optimized land use).

Target (80): The 2030 goal is to reach 80%. This signifies a shift from "sprawl" (lots of land with few services) to "compactness" (dense, serviced, and walkable land).

National vs. Regional Contribution

The index is not a flat national average; it is a weighted aggregate of 17 regional municipalities (Amanahs).

City-Level Weighting: Large cities like Riyadh, Jeddah, and Dammam have a higher weight in the national score because they represent the bulk of the urban population.

Regional Contributions: Each region has its own specific target based on its "Digital Twin" maturity. For example:

1. Riyadh might be aiming for 90% utilization due to the "Green Riyadh" and "Sports Boulevard" projects.
2. Smaller Regions might have lower initial targets as they transition from rural to urban zoning.

The Dashboard Logic: When your dashboard shows "58," it is pulling real-time data from Urban Observatories in each city. If a new park is opened in Medina, the "activated" land in that zone increases, which nudges the national index upward.

Map markers (13 zones across 6 regions):

#	Zone	Region	Area	Status
1	KAFD North Buffer Zone	Riyadh	12 KM²	CRITICAL
2	Diriyah Gate Western Corridor	Riyadh	8 KM²	CRITICAL
3	Eastern Industrial Fringe	Riyadh	15 KM²	HIGH
4	Al Awali Heights	Makkah	6 KM²	CRITICAL
5	Jeddah Al Hamdaniyah East	Makkah	10 KM²	HIGH

#	Zone	Region	Area	Status
6	Ras Al Khair Port Hinterland	Eastern	20 KM²	HIGH
7	Al Ahsa Southern Oasis Edge	Eastern	8 KM²	HIGH
8	Haramain Station District	Madinah	5 KM²	HIGH
9	Yanbu Al Sinaiyah Worker Zone	Madinah	7 KM²	HIGH
10	Abha City Center Infill	Asir	3 KM²	HIGH
11	Khamis Mushait Eastern Expansion	Asir	9 KM²	HIGH
12	Buraidah–Unaizah Corridor	Qassim	14 KM²	HIGH
13	Al Rass Heritage Quarter	Qassim	2 KM²	HIGH

Value to the Ministry

What the Ministry Gets	White Land Activation	Urban Utilization
Actionable intelligence	Specific plots with owner status, tax penalty amounts, and building permit history	Specific zones with km², why blocked, what to build
Policy enforcement	White Land Tax escalation decisions backed by satellite evidence	Rezoning and infrastructure priority decisions
Housing delivery	Direct link to Sakani waitlist — "this plot serves 320 families within 2 km"	"This zone can deliver 37,500 units at medium density"
Accountability	Track each plot from "idle" → "permit issued" → "construction started"	Track index score improvement per zone per quarter
Vision 2030 alignment	38% → 65% activation target with specific parcels mapped to close the gap	Score 58 → 80 with each zone's contribution quantified

1.4 Asset Evaluation Agent

> Analyzes site conditions to assess competitiveness and potential value of the area.

1.4.1 Infrastructure Competitiveness Index (`ast_1`)

Description: Scores zones 0–100 across transit, healthcare, education, utilities, green space, and digital infrastructure. Benchmarked against 50 global peer cities via MOMRA + GASTAT data.

Metric	Value
National Average (Q1 2026)	**54** / 100
2030 Target	**75** / 100 (Top 40 Global)
Gap	+21 pts
Zones Flagged	**11**

Calculation — 6 sub-dimensions (weighted):

Sub-Dimension	Weight	National Score	Target	Source

Sub-Dimension	Weight	National Score	Target	Source
🚌 Transit Access	20%	38	80	Population within 800m of metro/BRT/bus
⚡ Utility Reliability	20%	72	95	SEC grid uptime (SAIFI/SAIDI) + water supply
🏥 Healthcare Proximity	15%	52	80	Hospital beds per 1,000 + avg distance to primary care
🎓 Education Coverage	15%	58	85	School seats vs school-age population ratio
🌿 Green Space	15%	35	70	Public parks SQM per capita vs WHO standard (9 SQM/person)
📡 Digital Connectivity	15%	66	90	FTTH penetration + 5G coverage percentage

Data sources: MOMRA Municipal DB · GASTAT Infrastructure Census · SEC Reliability Indices (SAIFI/SAIDI) · MOH Facility Registry · MOE School Density · RCRC Green Space GIS · CITC Digital Coverage.

Map markers (11 zones, regional scores):

#	Zone	Region	Score	Severity
1	North Riyadh Metro Corridor	Riyadh	42/100	CRITICAL
2	South Riyadh Industrial Belt	Riyadh	51/100	HIGH
3	East Jeddah Residential Sprawl	Makkah	38/100	CRITICAL
4	Taif Mountain Gateway	Makkah	48/100	HIGH
5	Jubail Residential Sector	Eastern	56/100	HIGH
6	Al Khobar Waterfront District	Eastern	63/100	HIGH
7	Knowledge Economic City	Madinah	35/100	CRITICAL
8	Yanbu Heritage Coast	Madinah	47/100	HIGH
9	Abha City Center	Asir	52/100	HIGH
10	Buraidah Central District	Qassim	49/100	HIGH
11	Unaizah Heritage Core	Qassim	44/100	HIGH

Regional aggregated scores (from detail modal):

Region	Current	2030 Target	Gap
Riyadh	62	78	+16
Makkah	48	72	+24
Eastern	59	76	+17
Madinah	45	70	+25
Asir	52	71	+19
Qassim	47	68	+21

1.4.2 Real Estate Asset Yield Forecast (`ast_2`)

Description: ML model trained on 10-year REGA transaction data + satellite construction activity forecasts gross rental/investment yield per zone. Factors: demand (population + HQ relocations + tourism), supply (NHC pipeline), and giga-project proximity. Vision 2030 macro goal: Real estate sector contributes 10% of GDP (currently ~5.8%).

Metric	Value
Current National Avg Yield (Q1 2026)	**5.2%**
2030 Target Yield	**7.5%**
Gap	+2.3 percentage points
Zones Tracked	**9**
Vision 2030 Macro Goal	RE contributes 10% of GDP (*real)

Relationship between 7.5% yield target and 10% GDP goal: The 7.5% yield target is the investment attractiveness metric needed to attract sufficient domestic and foreign capital into Saudi real estate. Higher yields → more private investment → sector growth → achieving the macro 10% GDP contribution target.

Calculation formula:

```
Y_2030 = Y_NOW × D_MUL × (1 + P_GIGA) / S_PRESS × R_ADJ
```

5 yield factors (weighted):

Factor	Weight	Color	Description
D_MUL (Demand Multiplier)	30%	#FCD34D	Population growth + corporate HQ relocations + tourism influx
P_GIGA (Giga-Project Proximity)	25%	#00B558	Distance and phase-completion of nearby giga-projects (NEOM, KAFD, etc.)
S_PRESS (Supply Pressure)	20%	#ff4444	NHC housing pipeline + private developer supply compressing yields
R_ADJ (Risk Adjustment)	15%	#f97316	Macro risk, oil price correlation, regulatory stability
Y_NOW (Base Yield)	10%	#3b82f6	Current gross yield from REGA transaction registry

Data sources: REGA Transaction Registry (10-yr) · SAMA RE Finance Reports · NHC Housing Pipeline · STB Visitor Projections · GASTAT Population Census · Satellite Construction Detection (Sentinel-2).

Map markers (9 zones with yield trajectories):

#	Zone	Region	Current Yield	2030 Forecast	Confidence
1	KAFD–Diriyah Gate Corridor	Riyadh	6.1%	9.4%	82% (High)
2	East Riyadh Logistics Hub	Riyadh	4.8%	7.2%	71% (Mod-High)
3	Jeddah Central Waterfront	Makkah	5.4%	8.8%	76% (High)
4	Makkah Southern Residential	Makkah	4.5%	6.8%	68% (Moderate)
5	Dhahran Techno Valley	Eastern	5.8%	8.5%	79% (High)
6	Dammam Al Shatie District	Eastern	5.2%	7.0%	65% (Moderate)
7	Haramain Station District	Madinah	4.2%	7.1%	73% (Mod-High)
8	Soudah Peaks Tourism Zone	Asir	3.2%	6.5%	62% (Moderate)

#	Zone	Region	Current Yield	2030 Forecast	Confidence
9	Buraidah Agri-Tech District	Qassim	3.5%	5.8%	58% (Moderate)

Regional yield trajectories (from detail modal):

Region	Current	2030 Projected	Key Drivers	Confidence
Riyadh	5.8%	8.2%	KAFD, Diriyah Gate, corporate HQ	82%
Makkah	4.8%	7.5%	Jeddah Central, Umrah expansion	76%
Eastern	5.4%	7.8%	Aramco R&D, Techno Valley	79%
Madinah	4.2%	7.1%	Haramain station TOD, Umrah growth	73%
Asir	3.2%	6.5%	Soudah mega-project, domestic tourism	62%
Qassim	3.5%	5.8%	Agri-tech sector, NEOM Food Tech	58%

1.5 Citizen Insight Agent

> Chat based assistant answering questions about mobility patterns and citizen behaviors.

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2. Land Use Optimization and Zoning

Page route: `/optimization` **Full name:** Land Use Optimization and Zoning

This page optimizes land allocation, parking and accessibility, park coverage, project prioritization, and urban intervention potential — supporting balanced, livable urban development.

2.1 Zoning Mix Advisor Agent

> Optimizes land-use balance between residential, commercial, and services to support policy goals.

2.1.1 Land Use Balance Index (`zon_1`)

Description: Comprehensive score measuring deviation from Vision 2030 land allocation targets. AI analyzes zoning data to identify imbalances and suggests rezoning recommendations.

Metric	Value
Balance Score	**73** / 100
Trend	**+5** vs 2025

Target vs Actual land-use allocation:

Category	Target	Actual	Gap
Residential	35%	28%	-7%
Commercial	25%	34%	+9% (surplus)
Service	20%	17%	-3%
Green	20%	21%	+1%

AI Recommendation: Convert Al Olaya commercial zones to mixed-use residential. Projected score improvement: +12 points.

2.2 Accessibility and Parking Optimizer Agent

> Suggests optimal road, transport, and parking planning to improve accessibility.

2.2.1 Urban Accessibility Index (`prk_1`)

Description: Comprehensive index measuring urban transport efficiency and accessibility. AI analyzes parking turnover, public transit coverage, last-mile connectivity, and EV infrastructure to support Vision 2030 transportation goals.

Metric	Value
Efficiency Score	**72** / 100
Trend	**+3** vs 2025

Sub-dimensions:

Dimension	Weight	Current Score	Target
Parking Turnover	30%	65	100
Public Transit	25%	85	100
Last-Mile Connectivity	25%	73	100
EV Infrastructure	20%	64	100

2.2.2 EV Charging Density (`prk_2`)

Description: Predicts EV adoption rates to suggest optimal locations for charging hubs.

Metric	Value
Deficit Zones	**12**
Planned Stations	**45**

2.2.3 Last-Mile Connectivity (`prk_3`)

Description: Calculates the efficiency of micro-mobility links to major transit stations.

Metric	Value
Gaps	**6**
Efficient Links	**22**

2.3 Parks Selector Agent

> Ensures access to green spaces and parks, in line with goals and accessibility radius.

2.3.1 Green Space Coverage Index (`grn_1`)

Description: Comprehensive index measuring green space service capacity and accessibility. AI identifies optimal park locations to support Saudi Green Initiative and Vision 2030 sustainability targets.

Metric	Value
Coverage Score	**44** / 100
Trend	**+5** vs 2025

Metric	Value
Status	CRITICAL SHORTAGE

Sub-dimensions:

Dimension	Weight	Current Score	Actual Value
Per Capita Area	40%	11.3	1.7 m² (WHO standard: 9 m²) (*real — WHO recommended minimum)
Accessibility	30%	57.8	52% population coverage
Heat Relief	20%	70	2.1°C average cooling effect
Irrigation Efficiency	10%	81.3	65% smart irrigation

Metric	Value
New Parks Recommended	10
Total Area	60,000 m²
Service Population	60,000 people

2.3.2 Urban Heat Island Reduction (`grn_2`)

Description: Predicts temperature drops resulting from proposed canopy cover interventions.

Metric	Value
Severe Spots	**8**
Areas Cooled	**52**

2.3.3 Irrigation Efficiency (`grn_3`)

Description: Uses soil sensors and weather AI to minimize water use in public parks.

Metric	Value
Stressed Parks	**5**
Optimal Efficiency	**84%**

2.4 Priority Classifier Agent

> Assesses and classifies land, highlighting urgent priority zones for action.

2.4.1 Project Priority Index (`pri_1`)

Description: AI-powered priority ranking system for urban development projects. Evaluates bottleneck impact, time urgency, ROI, and synergy to optimize resource allocation and ensure Vision 2030 target achievement.

Metric	Value
P0 (Urgent) Projects	**1**
P1 (High) Projects	**2**
Backlog	**3**
Auto-Approval Rate	**75%**

Project priority rankings:

Priority	Project	Score	Status	Bottleneck	Urgency	ROI	Synergy	Delay	Action
P0	New Murabba	85	DELAYED	92	60	100	87	180 days	+25% WORKFORCE
P1	Riyadh Metro PH3	78	DELAYED	85	55	90	75	120 days	ENGINEERING REVIEW
P1	NEOM – The Line	72	AHEAD	45	70	95	82	On Track	PROCEED PHASE 2
P2	Diriyah Gate PH2	68	AHEAD	40	50	88	92	On Track	ACCELERATE TOURISM
P2	Red Sea Global PH2	65	ON TRACK	35	45	85	90	On Track	SUSTAIN MOMENTUM
P2	Qiddiya Entertainment	62	ON TRACK	30	40	90	85	On Track	ENTERTAINMENT FOCUS

2.4.2 Zoning Approval Speed (`pri_2`)

Description: Automates routine permit approvals while flagging complex cases for human review.

Metric	Value
Flagged (Manual Review)	**2**
Auto-Approved	**11K**

2.4.3 Infrastructure Urgency (`pri_3`)

Description: Ranks projects by impact on 2030 targets to optimize budget allocation.

Metric	Value
Critical Projects	**4**
Aligned Projects	**18**

2.5 Intervention Guide Agent

> Suggests urban interventions or upgrades to improve land use, accessibility, and service.

2.5.1 Urban Development Potential (`mob_1`)

Description: Comprehensive index measuring urban upgrade potential based on land use efficiency, accessibility, service coverage, and infrastructure. AI identifies priority intervention areas to support Vision 2030 population growth targets.

Metric	Value
Potential Score	**84** / 100
Status	PRIORITY UPGRADE
Demand Intensity	1.56×
Population Growth Target	150K → 250K

Sub-dimensions:

Dimension	Weight	Current	Target	Gap
Efficiency	30%	85	100	15
Accessibility	25%	73	100	27
Service Coverage	25%	75	100	25
Infrastructure	20%	98	100	2

Intervention priorities:

Priority	Action	Investment	Impact	ROI
P0	Traffic System Upgrade	500M SAR	-8 min commute	HIGH
P1	Service Facilities	300M SAR	+12% coverage	MEDIUM
P2	Land Use Optimization	100M SAR	+15% efficiency	MEDIUM

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3. Impact Simulation and Feasibility Assessment

Page route: `/simulation` **Full name:** Impact Simulation and Feasibility Assessment

This page runs "what-if" simulations for downtown Riyadh, testing traffic, mobility, planning trade-offs, financial viability, and environmental resilience under various parameter scenarios.

3.1 Urban Test Agent

> Runs simulations to test alternative zoning, road redesigns, or infrastructure proposals.

Indicators

ID	Indicator	Current	Target	Unit	Description
UT-1	Avg. Travel Speed	18	35	km/h	Average downtown travel speed during morning peak hours.
UT-2	Peak Hour Delay	45	15	min ↓	Average additional delay per trip during morning rush. Lower is better.
UT-3	Vehicle Throughput	12,400	18,000	veh/hr	Downtown corridor capacity during peak hours.
UT-4	CO_2 Reduction	0	40	%	Emission reduction from combined traffic optimization.

Simulation Settings

Setting	Slider Label	Range	Step	Default	Unit	Description

Setting	Slider Label	Range	Step	Default	Unit	Description
`additionalLanes`	Additional Lanes (Major Corridors)	0 – 4	1	0	lanes	Number of extra lanes added to 6 major downtown corridors (King Fahd Rd, Olaya St, King Abdullah Rd, Northern Ring, Makkah Rd, Eastern Ring). Methods: remove on-street parking (+1), road widening (+2), contraflow peak lanes (+1). Each lane adds ~800–1,000 veh/hr capacity per corridor.
`metroShift`	Metro Ridership Shift	0 – 40	2	0	%	Percentage of current car commuters shifting to Riyadh Metro Lines 1–6 (*real — 6 lines, 85 stations, 176 km, opened Dec 2024, world's longest driverless metro). Each 10% shift removes ~12,400 vehicles from downtown roads during peak hours.
`signalOptimization`	Signal Optimization Coverage	30 – 100	5	30	%	Percentage of downtown intersections upgraded with Adaptive Traffic Signal Control (ATSC). Baseline: 30% of 180 intersections already have SCATS/SCOOT systems (*real — real adaptive signal control technologies). Slider adjusts coverage up to 100%.
`congestionPricing`	Proposed Peak-Hour Toll	0 – 30	1	0	SAR	New congestion charge for vehicles entering the downtown core zone during peak hours (7–9 AM, 4–7 PM). Currently: 0 SAR (no toll exists). Reference: London ~70 SAR, Stockholm ~15 SAR, Singapore ~14 SAR (*real — approximate real-world congestion pricing benchmarks).

Computation Formulas

```
gain = (lanes / 4) × 0.30  +  (metroShift / 40) × 0.40
     + ((signalOpt − 30) / 70) × 0.25  +  (pricing / 30) × 0.15

Avg. Travel Speed   = min(42,  18 × (1 + gain))                    km/h
Peak Hour Delay     = max(8,   45 / (1 + gain))                    min
Vehicle Throughput  = min(22000,  12400 × (1 + lane_part + signal_part + metro_part))
veh/hr
CO₂ Reduction       = min(45,  metro_part + pricing_part + signal_part)
%
```

3.2 Mobility Impact Advisor Agent (Coming soon)

> Analyzes simulation results to recommend actions that improve accessibility and land-use.

Indicators

ID	Indicator	Current	Target	Unit	Description
MI-1	Pedestrian Mode Share	10.2	18	%	Suggests shade and cooling interventions to increase walking in desert climates.
MI-2	Active Transit Score	74	90	Score	Monitors bike-lane safety and usage to optimize future cycling infrastructure.
MI-3	Public Transit Accessibility	78	95	%	Ensures 80% of citizens live within 800m of a transport hub.

Simulation Settings

Setting	Slider Label	Range	Step	Default	Unit	Description
`shadeInfra`	Shade Infrastructure	0.5 – 3.0	0.1	1.0	×	Multiplier for shade and cooling interventions to increase walking in desert climates. Higher values represent increased shade canopy, misting systems, and covered walkways across pedestrian corridors.
`transitExpansion`	Transit Network Expansion	0.5 – 3.0	0.1	1.0	×	Multiplier for public transit network expansion scale. Higher values represent more metro extensions, bus routes, and transit hub coverage to bring more citizens within 800m of a transport stop.
`cyclingInvestment`	Cycling Investment	0.5 – 3.0	0.1	1.0	×	Multiplier for cycling infrastructure investment. Higher values represent more protected bike lanes, bike-sharing stations, and safety improvements to promote active transit.

Computation Formulas

```
Pedestrian Mode Share       = 10.2 × shadeInfra                          %
Active Transit Score        = 74 × (cyclingInvestment × 0.4 + 0.6)       Score
Public Transit Accessibility = 78 × (transitExpansion × 0.4 + 0.6)       %
```

3.3 Scenario Optimizer Agent (Coming soon)

> Compares alternatives and identifies optimal trade-offs between cost and sustainability.

Indicators

ID	Indicator	Current	Target	Unit	Description
SO-1	Sustainability Alignment	84	95	%	Ranks planning scenarios based on their carbon footprint and water efficiency.
SO-2	Cost-Benefit Ratio	3.2	4.5	Ratio	Analyzes long-term ROI of infrastructure vs. short-term construction costs.

ID	Indicator	Current	Target	Unit	Description
SO-3	Social Equity Score	88	100	Score	Ensures urban interventions are distributed fairly across all demographic groups.

Simulation Settings

Setting	Slider Label	Range	Step	Default	Unit	Description
`sustainabilityWeight`	Sustainability Weight	0.5 – 3.0	0.1	1.0	×	Multiplier weighting sustainability criteria (carbon footprint, water efficiency) in the scenario ranking model. Higher values prioritise greener outcomes over pure cost efficiency.
`investmentScale`	Investment Scale	0.5 – 3.0	0.1	1.0	×	Multiplier for overall infrastructure investment budget scale. Higher values increase CAPEX available for projects, improving cost-benefit ratios through economies of scale.
`equityPriority`	Equity Priority	0.5 – 3.0	0.1	1.0	×	Multiplier weighting equitable distribution of services across demographic groups. Higher values force the optimizer to favour socially balanced outcomes.

Computation Formulas

```
Sustainability Alignment = min(100,  84 × (sustainabilityWeight × 0.3 + 0.7))   %
Cost-Benefit Ratio       = 3.2 × (investmentScale × 0.4 + 0.6)                  Ratio
Social Equity Score      = min(100,  88 × (equityPriority × 0.3 + 0.7))         Score
```

3.4 Economic and Financial Analyzer Agent

> Models feasibility, funding needs, and economic return (e.g., self-sustaining neighborhoods).

Indicators

ID	Indicator	Current	Target	Unit	Description
EF-1	Project IRR	8	18	%	Internal Rate of Return for the neighbourhood development. Accounts for construction, land, infrastructure, and mixed-use revenue.
EF-2	Fiscal Self-Sufficiency	25	75	%	Percentage of neighbourhood costs covered by locally-generated revenue (property fees, rents, parking).
EF-3	Funding Gap	12	3	SAR B ↓	Unfunded infrastructure deficit. Reduced by PPP, land value capture, and developer contributions. Lower is better.
EF-4	Job-Housing Balance	0.6	1.2	Ratio	Ratio of jobs to housing units. 1.0 = equilibrium. Mixed-use and anchor tenants drive this up.

ID	Indicator	Current	Target	Unit	Description
EF-5	Private Investment Leverage	1.8	4.5	×	SAR private capital per SAR public. Higher = more attractive market conditions.

Simulation Settings

Setting	Slider Label	Range	Step	Default	Unit	Description
`mixedUseRatio`	Mixed-Use Development Ratio	15 – 60	5	15	%	Percentage of new development zones designated as mixed-use (retail + residential + office). Higher mixed-use creates walkable, self-sustaining neighbourhoods with diverse local revenue streams. Riyadh baseline ~15% reflects legacy single-use zoning.
`pppShare`	PPP Financing Share	10 – 70	5	20	%	Share of total infrastructure cost funded through Public-Private Partnerships. PPPs transfer construction and operational risk to private developers in exchange for revenue-sharing concessions (e.g., toll roads, district cooling). Reduces government fiscal burden and funding gap.
`farBonus`	FAR Bonus (Developer Incentive)	0 – 30	5	0	%	Floor Area Ratio bonus granted to developers who include public amenities (affordable units, parks, schools). Developers get extra buildable area; the city gets infrastructure contributions and density revenue without public spending.
`landValueCapture`	Land Value Capture Rate	0 – 40	5	5	%	Percentage of infrastructure-driven land value uplift recovered by the municipality through betterment levies, special assessment districts, or tax increment financing. When a metro station raises nearby land values 30–80%, LVC channels part of that windfall back into further development.
`anchorIncentive`	Anchor Tenant Incentive	0 – 50	5	0	%	Tax or rent concessions offered to attract major employers (corporate HQs, universities, hospitals) to anchor a neighbourhood. Creates employment gravity, drives foot traffic for surrounding retail, and attracts follow-on private investment.

Computation Formulas

```
# Normalize each gain to 0→1
mixedGain  = (mixedUseRatio - 15) / 45
pppGain    = (pppShare - 10) / 60
```

```
farGain    = farBonus / 30
lvcGain    = landValueCapture / 40
anchorGain = anchorIncentive / 50

Project IRR             = 8 + mixed×4.5 + ppp×2.5 + far×2.0 + lvc×0.5 + anchor×0.5
%
Fiscal Self-Sufficiency = min(85,  25 + mixed×25 + lvc×15 + far×8 + anchor×7)
%
Funding Gap             = max(1.5, 12 - ppp×4 - lvc×2.5 - far×1.5 - mixed×0.8)
SAR B
Job-Housing Balance     = min(1.4, 0.6 + mixed×0.3 + anchor×0.25 + far×0.1)
Ratio
Private Inv. Leverage   = min(5.5, 1.8 + ppp×1.2 + anchor×0.8 + mixed×0.4 + far×0.3)
×
```

Funding Sources Breakdown (Bar Chart)

The agent also renders a stacked horizontal bar chart showing funding composition across 4 sources:

Source	Color	Description
Government	#3b82f6	Direct public budget allocation
PPP	#10b981	Private sector via PPP concessions
LVC	#FCD34D	Municipal land value capture revenue
FAR/Dev	#f97316	Developer contributions via FAR bonuses

3.5 Environmental and Resilience Evaluator

> Assesses sustainability, and resilience to climate and flood risks.

Indicators

ID	Indicator	Current	Target	Unit	Description
ER-1	Flood Risk Index	42	85	Score	Composite score measuring flood resilience across wadi channels, basins, and 340km drain network.
ER-2	Heat Island Reduction	0	3.5	°C	Reduction in urban heat island effect from vegetation, reflective surfaces, and shade.
ER-3	Climate Resilience Score	35	90	%	Building stock meeting flood-proofing, heat-resistance, and water-efficiency standards. SBC baseline: 35%.

Simulation Settings

Setting	Slider Label	Range	Step	Default	Unit	Description
`drainageExpansion`	Drainage Infrastructure Expansion	0 – 50	5	0	km	Kilometres of new storm drain capacity added to Riyadh's existing 340km network. Targets wadi overflow zones (Hanifah, Al Aqiq) and low-lying urban basins. Each 10km reduces flood zone radius by ~12% in affected catchments.

Setting	Slider Label	Range	Step	Default	Unit	Description
`retentionBasins`	Retention Basins & Flood Barriers	0 – 8	1	0	facilities	Number of new retention basins and flood barrier facilities constructed at key flood-prone nodes (wadi confluences, highway underpasses, district outfall points). Each facility stores ~50,000m³ of stormwater runoff during peak events.
`greenCover`	Urban Green Cover	8 – 30	1	8	%	Percentage of city area under vegetation canopy (parks, street trees, green corridors). Baseline 8% is among the lowest globally for a major city. Each 1% increase reduces local surface temperature by ~0.15°C and improves stormwater absorption.
`coolRoof`	Cool Roof & Reflective Surface Coverage	5 – 60	5	5	%	Percentage of rooftops and paved surfaces treated with high-albedo (reflective) coatings or cool-roof materials. Reflects solar radiation instead of absorbing it, reducing building cooling loads by 10–20% and ambient air temperature by up to 2°C in dense districts.
`buildingCode`	Climate-Resilient Building Code	35 – 100	5	35	%	Adoption rate of updated Saudi Building Code (SBC 601/602) standards for flood-proofing, heat resistance, and water efficiency in new construction. Baseline ~35% reflects current MOMRA enforcement. Higher adoption reduces structural flood damage and cooling energy demand.

Computation Formulas

```
drainGain = drainageExpansion / 50
basinGain = retentionBasins / 8
greenGain = (greenCover − 8) / 22
codeGain  = (buildingCode − 35) / 65
coolGain  = (coolRoof − 5) / 55

Flood Risk Index       = min(95,   42 + drainGain×28 + basinGain×18 + codeGain×7)
Score
Heat Island Reduction  = min(5.5,  greenGain×2.5 + coolGain×1.8 + codeGain×0.6)
°C
Climate Resilience Score = min(95, 35 + codeGain×30 + drainGain×12 + basinGain×8 +
greenGain×5 + coolGain×5)  %
```

Flood Spot Simulation

Each flood spot's radius and risk score react to parameter changes:

```
drainR = 1 - (drainageExpansion / 50) × 0.35
basinR = 1 - (retentionBasins / 8) × 0.40
codeR  = 1 - ((buildingCode - 35) / 65) × 0.15

simRadius = max(50, baseRadius × drainR × basinR × codeR)
simRisk   = max(8,  risk × drainR × basinR × codeR)
```

Heat Island Simulation

Heat island intensity per point reacts to green cover, cool roofs, and building codes:

```
greenR = 1 - ((greenCover - 8) / 22) × 0.35
coolR  = 1 - ((coolRoof - 5) / 55) × 0.30
codeR  = 1 - ((buildingCode - 35) / 65) × 0.15

simWeight = max(0.02,  baseWeight × (isUrban ? greenR × coolR : 1) × codeR)
```

Flood Risk Spot Markers (7 locations in Riyadh)

Location	Base Drainage Radius	Risk Score
Wadi Hanifah — Al Diriyah	850m	92
Wadi Hanifah — Al Aqiq	750m	88
King Fahd Rd Underpass	400m	85
South Basin — Exit 15	1,200m	78
Olaya Depression	350m	72
Northern Ring Catchment	600m	68
Eastern District Basin	900m	65

---

4. Monitoring and Improvement

Page route: `/act4` **Full name:** Monitoring and Improvement

This page provides continuous monitoring of urban change, asset utilization, and development pipeline progress — using satellite/LiDAR AI to validate ground truth.

4.1 Urban Monitoring Module (Change Tracker Agent)

> Tracks spatial and development changes using geospatial and data to update the digital twin.

Spatial Truth KPIs:

Metric	Value
Masterplan Approved Allocation	**500 KM²**
Footprint Drift (Deviation)	**+2.8%**
Urban Sprawl Discrepancy	**14.2 KM²** (LiDAR truth vs Amanah reports)
Financial Exposure	**2.4B SAR** (est. infrastructure cost if developed)

Urban sprawl trend data (Jan–Dec):

Reported (Amanah): 490 → 500 KM² Detected (LiDAR): 490 → 514.2 KM²

Land use transformation (2017–2026):

Year	Residential	Commercial	Green	Industrial	Total (KM²)
2017	438	89	9.0	125	661
2020	505	108	11.2	148	772
2023	595	134	13.5	175	918
2026 (proj.)	705	172	15.0	218	1,110

Green space by region:

Region	Current (KM²)	Target	Growth	Status
Riyadh	12.8	15.0	+0.6	On Track
Jeddah	11.5	15.0	+0.5	Attention
Mecca	14.2	15.0	+0.8	Exceeding
Medina	13.8	15.0	+0.7	On Track
Dammam	10.8	15.0	+0.4	Critical
Tabuk	15.5	15.0	+1.2	Exceeding

4.2 Condition Watch Agent

> Monitors utilization and physical conditions of parking and infrastructure assets.

CAPEX Utilization — Projects by idle rate:

Project	Code	Idle Rate	Budget (B SAR)	Spent (B SAR)	Recommendation
Riyadh Metro Extension PH2	RME-PH2	78.5%	4.2	1.8	DEFER
Jeddah Waterfront Dev PH3	JWD-PH3	68.4%	3.8	2.1	REVIEW
Dammam Industrial Zone	DIZ-01	52.3%	2.5	1.9	PROCEED
Mecca Transport Hub	MTH-02	41.8%	5.1	4.2	PROCEED
Medina Smart City PH1	MSC-PH1	28.6%	1.8	1.5	PROCEED
Tabuk Renewable Energy	TRE-01	15.2%	1.2	0.9	PROCEED

Population pressure trend (quarterly):

Quarter	Influx Index	Capacity
24 Q1	75	100
25 Q1	86	100
26 Q1	97	100
27 Q1 (proj.)	108	100 ⚠ OVER CAPACITY

4.3 Development Pipeline (Change Tracker Agent)

> Monitors giga-project progress using satellite, LiDAR and AI detection.

Progress calculation logic:

```
Planned Progress = (Current Year - Start Year + 1) / (Deadline Year - Start Year + 1) ×
100
AHEAD:   Actual > Planned + 5%
DELAYED: Actual < Planned - 5%
ON TRACK: Within ±5% of planned
```

Projects tracked:

Project	Type	Location	Total Area	Progress	Budget (B SAR)	Deadline	Status
NEOM – The Line	Mega City	Tabuk (*real)	170 KM	72.4%	500	2030 (*real)	AHEAD (+12.4%)
Diriyah Gate PH2	Heritage	Riyadh (*real)	22 KM²	86.4%	45	2028	AHEAD (+15%)
New Murabba	Downtown	Riyadh (*real)	19 KM² (*real)	26.3%	104	2030 (*real)	DELAYED (-33.7%)
Riyadh Metro PH3	Infrastructure	Riyadh (*real)	45 KM	33.3%	28	2029	DELAYED (-29.2%)
Red Sea Global PH2	Tourism	Red Sea (*real)	35 KM²	74.3%	32	2028	ON TRACK (+2.9%)
Qiddiya Entertainment	Entertainment	Riyadh (*real)	366 KM² (*real — Wikipedia cites ~334 KM²)	76.5%	75	2028	ON TRACK (+1.5%)

> **Note on giga-project data:** Project names, types, locations, and deadlines are **real** (sourced from official announcements). Total areas are approximately real. However, progress percentages, budget figures (B SAR), and status labels (AHEAD/DELAYED/ON TRACK) are **sample data** fabricated for the demo.

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5. Panorama (Home Page)

Page route: `/` (index) **Full name:** Panorama

Overview dashboard providing at-a-glance KPIs across all four functional areas, with a Saudi Arabia macro map view.

Summary KPIs shown on Panorama

Section	KPI Card	Metric	Unit	Description
Diagnostics	Flow Agent (24H Commute)	Chart	—	Line chart with >45M threshold
Diagnostics	Demand Forecaster	-153K	UNITS	Housing Deficit by 2030
Diagnostics	Idle Land Agent	38	%	White Land Activation

Section	KPI Card	Metric	Unit	Description
Optimization	Zoning Advisor	35%	—	Over-limit (pie chart center)
Optimization	Intervention Guide	47	%	Optimal ROI
Optimization	Access & Parking	13	ZONES	Strain <5%
Simulation	Flood Risk Index	42	Score	Current · 2030 Target: 85
Simulation	Project IRR	8%	%	Current · 2030 Target: 18%
Monitoring	Change Tracker	12	SITES	Sat-AI Violations
Monitoring	Cont. Learning	+8.3	%	Model Accuracy Lift
Monitoring	Sync Delay	Chart	—	Bar chart with >1H threshold

Map markers on Panorama

Label	Metric	Description	Type
WHITE LAND	4 YRS	Al Yasmin Untaxed	Warning
ZONING STRAIN	35%	KAFD Commercial Overload	Alert
CRITICAL FLOW	12 KM/H	King Fahd Speed Drop	Alert
FLOOD RISK	98%	Wadi Hanifah Load	Warning

---

6. Data Authenticity Reference

> Classification of all data in this demo as **REAL** (factual/verifiable) or **SAMPLE** (fabricated for demonstration purposes).

6.1 Real / Factual Data

These items are based on verifiable public information, cross-referenced against government websites and reliable third-party sources (March 2026):

6.1.1 Previously Listed (structural references)

Category	Examples	Source
Vision 2030 targets	70% homeownership, RE 10% of GDP	Saudi Vision 2030 official documents
Saudi geography	City names, region names, wadi names (Wadi Hanifah, Al Aqiq)	Standard geographic references
Government agencies	MOMRAH, NHC, GASTAT, REGA, SAMA, SEC, MOH, MOE, CITC, RCRC, STB	Real Saudi government agencies
Giga-project names	NEOM – The Line, Diriyah Gate, New Murabba, Red Sea Global, Qiddiya, KAFD	Real Saudi giga-projects
Standards & frameworks	WHO 9 m²/person green space, Saudi Building Code (SBC 601/602), SAIFI/SAIDI reliability indices	Real international/national standards
Infrastructure references	Riyadh Metro Lines 1–6, SCATS/SCOOT signal systems, White Land Tax (2.5% levy)	Real infrastructure programs/policies

Category	Examples	Source
Riyadh road names	King Fahd Rd, Olaya St, King Abdullah Rd, Makkah Rd, Northern Ring, Eastern Ring, etc.	Real street names
Benchmark references	London congestion charge (~70 SAR), Stockholm (~15 SAR), Singapore (~14 SAR)	Approximate real-world congestion pricing
Riyadh existing drain network	340km baseline network	Approximate real figure

6.1.2 Verified Targets and Values (online-confirmed)

The following specific numbers used in the demo have been verified against official/reliable sources:

Data Point	Value in Demo	Verified Value	Source	Section
Homeownership 2030 target	70%	70% ✅	Vision 2030 official target; confirmed by Global Property Guide, multiple news outlets	`dmd_1`
White Land Tax rate	2.5% annual levy	2.5% ✅	Saudi Cabinet-approved policy (plots >5,000 sqm); confirmed by Global Property Guide, JLL reports	`idl_1`
WHO green space standard	9 m²/person	9 m²/person ✅	WHO recommended minimum for urban green space	`grn_1`
RE sector 10% of GDP target	10% of GDP	10% of GDP ✅	Vision 2030 macro goal for real estate sector contribution	`ast_2`
Riyadh Metro	Lines 1–6	6 lines, 85 stations, 176 km ✅	Opened Dec 2024; Guinness World Record longest driverless metro (Wikipedia, RCRC)	Sec 3.1
SCATS/SCOOT systems	Referenced as existing signal tech	Real adaptive traffic control systems ✅	Used globally including Saudi Arabia	Sec 3.1
Congestion pricing benchmarks	London ~70 SAR, Stockholm ~15 SAR, Singapore ~14 SAR	Approximately correct ✅	London ~£15/day ≈ 67 SAR; Stockholm ~SEK 60 ≈ 15 SAR; Singapore ~S$5 ≈ 14 SAR	Sec 3.1
Saudi Building Code	SBC 601/602	Real code references ✅	Saudi MOMRA building code standards	`ast_1`, Sec 3.5
SAIFI/SAIDI indices	Referenced as utility reliability metrics	Standard IEEE power reliability indices ✅	Used by SEC (Saudi Electricity Company)	`ast_1`
New Murabba area	19 KM²	19 KM² ✅	Confirmed by Wikipedia, MEED, official project announcement	Sec 4.3
New Murabba deadline	2030	2030 ✅	Official project timeline (Construction Week)	Sec 4.3
NEOM location	Tabuk	Tabuk ✅	Official NEOM announcement	Sec 4.3
NEOM deadline	2030	2030 (first phase) ✅	Full completion expected ~2045; first 5 km segment by 2030	Sec 4.3

Data Point	Value in Demo	Verified Value	Source	Section
Qiddiya area	366 KM²	~334 KM² (≈ close)	Wikipedia / official sources cite 334 sq km	Sec 4.3
Sakani housing program	Referenced as NHC initiative	Real program ✅	Launched 2018; >800,000 contracts facilitated (Vision 2030 website)	`dmd_1`
Saudi population	~8.6M Riyadh (regional)	Total national: ~35.3M ✅	GASTAT 2024 census data	`dmd_1`
Homeownership baseline (2016)	—	~47% in 2016 ✅	Global Property Guide citing GASTAT	—
Homeownership (actual, early 2025)	63% (Q1 2026 in demo)	~65.4% (early 2025) ⚠	Demo uses 63%, real figure was 63.7% in 2024, 65.4% by early 2025	`dmd_1`

> **Note:** The homeownership rate of 63% used in the demo is close to the real 2024 figure (63.7% per GASTAT/Global Property Guide). By Q1 2026, the actual rate was likely ~65–66%. The demo value is directionally correct but slightly outdated.

6.2 Sample / Dummy Data

All numerical values, indicators, map markers, project statuses, and simulation results are **fabricated for demo purposes** unless specifically marked (*real) above:

Category	What is sample	Notes
All indicator values	78/100 commute index, 38% activation, 54/100 infra score, 5.2% yield, etc.	Illustrative, not from real datasets
Most 2030 target values	92/100, 65%, 75/100, 7.5%, etc.	Realistic but not official per-indicator targets. **Exceptions:** 70% homeownership and RE 10% of GDP are (*real) Vision 2030 targets
Map marker locations	Al Yasmin Block 4A, Al Malqa Apex, KAFD North Buffer Zone, etc.	Location names are plausible but parcels/zones are fictional
Map marker data	Areas (14,500 SQM), idle years (4 yrs), severity levels, risk scores	All fabricated
City congestion rankings	Riyadh 78, Jeddah 72, Makkah 68, etc.	Fictional index values
Road project data	Riyadh–NEOM Corridor 4,200 KM, Jeddah–KAEC Expressway 1,800 KM, etc.	Fictional projects and lengths
Housing deficit figures	-153K total, -35K Riyadh, -42K Makkah, etc.	Illustrative, not from NHC data
Regional yield data	5.8% Riyadh, 4.8% Makkah, zone-level forecasts, confidence %	All fabricated

Category	What is sample	Notes
Agent/zone scores	All sub-dimension scores, weighted calculations	Designed to look realistic but are dummy
Project pipeline statuses	NEOM 72.4%, New Murabba 26.3%, budget figures (500B, 104B, etc.)	Progress percentages and budget figures are fictional. **Project names, types, locations, areas (e.g. New Murabba 19 KM²), and deadlines are (real)*
Monitoring data	Footprint drift +2.8%, urban sprawl 14.2 KM², CAPEX idle rates	All sample data
Simulation formulas	UT gain, MI multipliers, EF financial model, ER flood model	Simplified models for demo; not calibrated to real-world data
Simulation defaults & ranges	0–4 lanes, 0–40% metro shift, 15–60% mixed-use, etc.	Plausible ranges but not from any urban planning study
Flood spot base radii & risk scores	850m/92, 750m/88, etc.	Fictional baseline measurements
Heat island point weights	Per-point weights and urban flags	Fabricated for visualisation
Funding source breakdowns	Government vs PPP vs LVC vs FAR allocation bars	Computed from dummy formula, not real budgets
Population/homeownership by region	8.6M Riyadh, 55% ownership, etc.	Approximate but not official GASTAT figures

6.3 Summary Rule

> **If it is a number, score, percentage, area, budget, deficit, index, rank, progress, or map coordinate in this demo — it is SAMPLE data, UNLESS marked with (real).*
>
> Confirmed (*real) values: **70% homeownership target**, **White Land Tax 2.5%**, **WHO 9 m²/person**, **RE 10% of GDP target**, **Riyadh Metro 6 lines / 176 km**, **New Murabba 19 KM² / 2030 deadline**, **NEOM Tabuk / 2030 deadline**, **SCATS/SCOOT systems**, **SBC 601/602**, **SAIFI/SAIDI**, **congestion pricing benchmarks**.
>
> All other numerical values (scores, indices, deficits, yields, progress %, budget figures, map marker data) are SAMPLE.
>
> Structural references (agency names, city names, policy names, project names, standards) are real.

Appendix: Agent Index by Page

#	Page	Agent Name	Description
1	Diagnostics	**Flow Agent**	Forecasts citizen movement patterns and changing demand for services
2	Diagnostics	**Demand Forecaster Agent**	Forecasts future needs for housing, roads, and amenities across regions
3	Diagnostics	**Idle Land Agent**	Identifies and maps underdeveloped or vacant parcels for redevelopment potential

#	Page	Agent Name	Description
4	Diagnostics	**Asset Evaluation Agent**	Analyzes site conditions to assess competitiveness and potential value of the area
5	Diagnostics	**Citizen Insight Agent**	Chat based assistant answering questions about mobility patterns and citizen behaviors. Powered by HiAgents
6	Optimization	**Zoning Mix Advisor Agent**	Optimizes land-use balance between residential, commercial, and services to support policy goals
7	Optimization	**Accessibility and Parking Optimizer Agent**	Suggests optimal road, transport, and parking planning to improve accessibility
8	Optimization	**Parks Selector Agent**	Ensures access to green spaces and parks, in line with goals and accessibility radius
9	Optimization	**Priority Classifier Agent**	Assesses and classifies land, highlighting urgent priority zones for action
10	Optimization	**Intervention Guide Agent**	Suggests urban interventions or upgrades to improve land use, accessibility, and service
11	Simulation	**Urban Test Agent**	Runs simulations to test alternative zoning, road redesigns, or infrastructure proposals
12	Simulation	**Mobility Impact Advisor Agent**	Analyzes simulation results to recommend actions that improve accessibility and land-use
13	Simulation	**Scenario Optimizer Agent**	Compares alternatives and identifies optimal trade-offs between cost and sustainability
14	Simulation	**Economic and Financial Analyzer Agent**	Models feasibility, funding needs, and economic return (e.g., self-sustaining neighborhoods)
15	Simulation	**Environmental and Resilience Evaluator**	Assesses sustainability, and resilience to climate and flood risks
16	Monitoring	**Urban Monitoring Module**	Tracks spatial and development changes using geospatial data to update the digital twin
17	Monitoring	**Condition Watch Agent**	Monitors utilization and physical conditions of parking and infrastructure assets
18	Monitoring	**Change Tracker Agent**	Detects and monitors spatial/urban land-use changes
19	Monitoring	**Continuous Learning Agent**	Ensures continuous learning and adjustments based on changes in policies, demographics, etc.